Layer 5

Request Life cycle:

when Request hits LLM, it goes through a strict sequence of operation:

Request arrive → input tokenized → prompt is built or chat template

→ prefill Phase occurs → First token generated → decode loop begin

→ output stream back to the user → Stop condition is met

Logged

Prefill vs Decode

prefill: processing input prompt to create initial KV's → TTFT (time to first token) a must to watch

↑ prompts ↑ TTFT

Decode : Generate output sequentialy, one token at a time
TPot (time per output token).

constrained by :
- → memo bandwith
- → Speed of Read KV cache

Latency vs Throughput

how long user will actually waits ← Latency

a system may achive high throughput but poor user latency

Throughput → measures token per sec

Batching stratigies : effciently utilize GPU's

— static batching : LLM Requests have highly variable lengths → wasting hardware capacity

= continous batching : dynamic approach that → add
→ Removes

Requests on the Fly

↳ improve utilization when handiling live trafic

essiential metrics to track : TTFT, TPoT, Latency, throughput (P50, P95, P99) Latency percentiles, GPU and RAM utilization and usages Queue time, error Rate

END OF inFerance Fundamentals